

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Based Code Similarity and Plagiarism Explanation System

Abstract

The increasing adoption of online programming platforms, digital learning environments, and automated programming assessments has made source code plagiarism a significant challenge in academic institutions. Conventional plagiarism detection techniques primarily rely on textual or token-level comparison and may not effectively identify modified source code in which variable names, comments, formatting, or code organization have been changed while preserving similar programming logic. This project proposes an **AI-Based Code Similarity and Plagiarism Explanation System** that combines source code preprocessing, lexical analysis, structural analysis, and semantic code representation to identify and explain similarities between programming submissions. The system provides a web-based platform through which faculty can create programming assignments and students can submit source code for automated analysis. Submitted programs are preprocessed through comment removal, formatting normalization, tokenization, and identifier handling, followed by similarity analysis using TF-IDF with Cosine Similarity, token-based comparison, and Abstract Syntax Tree (AST)-based structural analysis. To enhance semantic similarity detection, pretrained code representations such as CodeBERT are incorporated to capture meaningful relationships between source code segments beyond exact textual matches. The outputs from these analysis techniques are combined to generate an overall similarity score and a similarity classification that assists faculty in identifying potentially suspicious submissions. An explainable analysis module further highlights corresponding code regions and provides explanations for detected similarities, including common statements, control structures, equivalent operations, and identifier modifications. The system also provides a dashboard for assignment and submission management, pairwise code comparison, similarity visualization, analysis history, faculty review, and automated plagiarism report generation. The proposed system aims to provide an intelligent, transparent, and efficient solution for source code similarity analysis by integrating artificial intelligence with program structure and code analysis techniques, thereby supporting academic integrity and reducing the manual effort involved in reviewing programming submissions.

MAJOR PROJECT REGISTRATION FORM

Date:

References (Category specific)

1. Fahad Ebrahim, Mike Joy, "Source Code Plagiarism Detection with Pre-Trained Model Embeddings and Automated Machine Learning," Proceedings of the 14th International Conference on Recent Advances in Natural Language Processing (RANLP), 2023, pp. 301–309.
2. Nandini Gandhi, Kaushik Gopalan, Prajish Prasad, "A Support Vector Machine Based Approach for Plagiarism Detection in Python Code Submissions in Undergraduate Settings," Frontiers in Computer Science, Vol. 6, 2024, DOI: 10.3389/fcomp.2024.1393723.
3. Fahad Ebrahim, Mike Joy, "Semantic Similarity Search for Source Code Plagiarism Detection: An Exploratory Study," Proceedings of the ACM ITiCSE Conference, 2024, pp. 360–366, DOI: 10.1145/3649217.3653622.
4. Jorge Martinez-Gil, "Improving Source Code Similarity Detection Through GraphCodeBERT and Integration of Additional Features," arXiv Preprint, 2024.

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A4250	Gavini Keerthi	
2	23BQ1A4238	Derangula VishnuVardhan	
3	23BQ1A4255	Gorla Neelima	
4	23BQ1A4230	Chimata Gireesh Babu	

Project Guide

Name: Mrs. K. Deepika

Signature:

MAJOR PROJECT REGISTRATION FORM

Date:

INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.